Native Hyperdimensional Attention, Made Small

A controlled study of GHRR phase-binding (arXiv:2405.09689) inside a 36 M-parameter language model — gate-by-gate, from first oracle check to a fine-tuned null

Experimental Agentic AI under human guidance · ScivizLab, University of Applied Arts Vienna
Code and paper: `github.com/ScivizLabVienna/nativehdc` (release v1.0; archival DOI via Zenodo)
Autonomous research log under operator-set gates · revised 6 November 2026 · experiments G0–G10 complete

Abstract.

We re-implemented generalized holographic reduced representations (GHRR) attention from scratch as a drop-in softmax replacement and asked two questions under pre-registered gates: can it train stably end-to-end inside a standard Transformer block, and whether the mechanism yields measurable perplexity benefit on a standard corpus. The answer to the first is affirmative: our module matches the HoloVec oracle to 2.4e-7 on binding and the hash encodings reproduce across interpreter restarts without any hash-seed patching tricks beyond one documented rule. The second produced a qualified keep: at matched budget GHRR lost the bake-off by 12.7 %, closed to +8.3 % with one learning-rate correction, and stayed inside the ±10 % margin we had fixed in advance. Two follow-up ablations beat the baseline instead: sharpening the binding temperature (τ 4 → 16 over two promotions) gained 2.7 % validation perplexity, and — our main result — cutting the hypervector dimension from 4096 to 1024 cut parameters by 44.2 % (64.2 M → 35.9 M) *and* improved perplexity (5.853 → 5.831), with all associative-memory gates passing. A capacity experiment shows why a floor exists: facts bundled into one D=4096 vector saturate at ~246 items, matching HDC theory to within noise. On ProofWriter rule chains, both architectures sit in the same majority-class regime zero-shot (0.271 vs 0.266) and stay statistically indistinguishable after fine-tuning (0.501 vs 0.499, flat across inference depth). The negative results are reported as negatives; a four-arm seed-variance study detailed in Section 5.3, confirms the dimension-shrink advantage across paired seeds but qualifies its magnitude as equal-or-marginal-better at strictly lower cost.

1 Why this experiment exists

Hyperdimensional computing proposes that symbolic structure — binding, bundling, similarity lookup — can be carried by ordinary differentiable tensors rather than maintained as separate symbolic machinery. The open question at small model scale is whether *inside* attention and train the whole thing with backpropagation, does it actually help a language model, or is it an expensive way to re-derive a positional prior?

We set out to answer that for one concrete published mechanism — GHRR (C. Yeung, Z. Zou, S. Jeong, W. Huang, N. Bastian & M. Imani, arXiv:2405.09689v2, IEEE TAI 2026), which represents HVs as unitary matrices and shows an algebraic equivalence between GHRR binding and a restricted form of attention — rather than inventing our own variant. Everything below runs against criteria fixed before measurement: pre-registered accept/reject margins, one authorized retry per failure mode, and a recorded verdict line for every gate. The paper reports each gate outcome, including two results contrary to expectations drawn from the source literature.

Three claims this log can defend:

- **GHRR attention trains stably end-to-end** in a 36–64 M parameter LM (181.8 M-token TinyStories budget, RTX 3090) — oracle-verified numerics, finite gradients everywhere,

no NaN across ~50 GPU-hours of runs.

- **The binding score temperature is a first-class hyperparameter**: cumulative −2.7 % validation perplexity from τ=4 to the τ=16 champion, realized in two paired replications (−1.9 % then −0.7 %), with diminishing returns after τ≈8.
- **Fourfold reduction of hypervector dimension reduces cost without loss of quality:** D=1024 beats D=4096 at full training budget while using 44.2 % fewer parameters — provided the memory-workload design keeps per-vector bundle counts far below the theoretical ~D/(2·ln D) ≈ 246-item saturation point.

2 Background: what GHRR actually says

In GHRR a hypervector of dimension *D* and multiplicity *m* is a block matrix A = QΛ with Q a fixed random unitary basis and Λ the diagonal of i.i.d. uniform phases. Binding is matrix multiplication; similarity is the normalized real trace inner product; the quasi-orthogonality that makes HDC work follows from E[ηλ*] = 0 for independent phases (their Proposition 1). Section IV-E rewrites attention as a GHRR convolution — [σ(Q*K†)*V] restricted per head — which is the property we exploit: replace softmax(qk^T) with a phase-rotation score and leave all other components unchanged. The paper itself permits relaxing unitarity of the trained maps “for expressivity comparable to a traditional Transformer,” so our trainable projections stay plain linear layers.

Two reading details shaped the design and are worth stating because they are easy to misread. First, in the paper's own semantics **D counts heads and m counts independently handled tokens** — their headline system uses small D with m ≈ context length. Our first implementation is deliberately the degenerate m = 1 path: one FHRR-style phase rotation per token slot, scalar phase tables (a full-D phase tensor over a 50k vocabulary was memory-infeasible), and softmax attention otherwise unchanged. Whatever our numbers show, they show about *phase-kernel positional binding*, not about the paper's flagship multi-binding regime. Second, their language-model result (§V-E) is itself modest: Wikitext2 perplexity 29.16 → 27.54 (−5.5 %, mean of five runs) on small models with fixed random position encodings that may vary across Q/K/V and heads. Our ±10 % accept margin was calibrated to expect exactly this neighborhood.

3 System, verification gates G0–G1

Python 3.13 · PyTorch 2.14.1+cu126 · single RTX 3090 (24 GB) · CUDA via uv-managed venv. The LM (`TinyLM`) is a stock decoder-only Transformer: d_model = 384, 6 layers, 8 heads, tied embeddings, context 64, bf16 — identical in both arms except the attention kernel. Table 1 states the configuration that produced our champion.

Setting	Champion (G9, promoted 2026-11-06)
Attention	GHRR phase kernel, m=1, τ=16
Hypervector dim D	1024 (was 4096 before gate G6)
Parameters	35.9 M (64.2 M at D=4096; −44.2 %)
Optimizer	Adam-family, cosine schedule, peak LR 3e-4, warmup 500
Data	TinyStories, gpt2 BPE, ≤800-char docs; 181.8 M unique tokens, ~1.65 epochs
Batch / ctx	16 × 64, eval every 1000 steps, best-on-val checkpoints
Seed	1337 (replay verified bit-deterministic; G8 adds seed sweep)
Inference env contract	GHRR_TAU=8 GHRR_HV=1024 required — scalars do not serialize in state dicts; strict load fails loudly

Table 1 — champion configuration; anything a checkpoint does not carry is stated here.

G0 (environment reproduction). We first re-ran the mechanism the way its own reference implementation would: GHRR(dim=256, m=3) binding round-trips at similarity 1.000000, non-commutativity of bind shows up as sim(ab, ba) = 0.16, and the m=1 restriction commutes exactly (the FHRR limit), again 1.000000. Exit code zero on first try.

G1 (our module vs HoloVec oracle). Our hand-written FHRR kernel was scored against HoloVec's reference operations: binding error 2.4e-7, similarity error 1.0e-8 — both float-noise. The blake2b hash encodings come out quasi-orthogonal ($|\text{sim}| \approx 0.009 <$ the $3/\sqrt{D}$ statistical bound) and byte-identical across processes with different PYTHONHASHSEED, which is only true because we hash an explicit digest rather than Python's randomized builtin — a trap for anyone tempted to "simplify" the encoder. The attention module passed its causal-mask leak test, its self-score identity (diagonal scores equal τ exactly), and a full-gradient-finiteness sweep.

4 Protocol: pre-registered gates against one corpus

The bake-off corpus is TinyStories truncated to 800-character documents. Two deviations are recorded rather than concealed: the ≤800-char filter exhausted the stream at **181.8 M unique tokens**, short of the 300 M target, so both arms (and every ablation) trained on the identical corpus, raising effective epochs to ~1.65 — perplexities are comparable *within* this protocol and nowhere else; and the vanilla arm was tuned only once (the original bake-off LR), a fairness note restated in every gate comparison so the baseline cannot drift.

Gate discipline: each experiment declares its verdict rule before launch — e.g. G2 keeps GHRR iff it lands within ±10 % of vanilla's achieved perplexity, with exactly one authorized learning-rate retry on failure; G6 promotes a new dimension only if the full-budget retrain beats the champion; G7 requires a >1 percentage-point accuracy delta plus a shallower depth-decay slope to claim a reasoning advantage. Every verdict line lands in `STATUS.md` with command, numbers, and date.

5 Results gate by gate

5.1 G2 — the bake-off, twice

Run 1 (both arms, LR 6e-4): vanilla 5.511, GHRR 6.211 — a 12.7 % loss, outside the pre-registered ±10 %, with the per-step gap flat at +13–15 % from step 1k onward and GHRR costing ~1.9× wall-clock at D=4096. The failure rule authorized one LR retry; we chose a *lower* peak (3e-4) because the curves at 6e-4 were smooth, which argues for an optimization mismatch on the phase tables rather than instability.

Run 2 confirmed the diagnosis. GHRR@3e-4 reached **5.966**, +8.3 % versus retained vanilla — inside ±10 % → verdict **KEEP GHRR**, emitted mechanically by the report script. The corroborating detail matters: at 3e-4 the gap *shrank monotonically from +31 % to +7–8 %* as budget grew, while at 6e-4 it never moved. A capacity ceiling would not do that; a too-hot optimizer on a new parameter geometry does.

5.2 G5 — sharpening the binding score

The τ knob controls how peaked phase-similarity is allowed to become before it acts as an attention weight. Six short probes (τ ∈ {2,4,8,16} × context 64/128) at equal step budgets: improvement is strictly monotone in τ up to 8 (−9 %, then −5–7 % per doubling), and the two-point follow-up at τ=16 shows gains collapsing to ~1–2 %, indicating saturation rather than a monotone frontier. Promoting τ=8 into a full-budget retrain realized −1.9 % (5.966 → **5.853**) and the champion was promoted on that number after re-verifying both downstream gates at the new scalar.

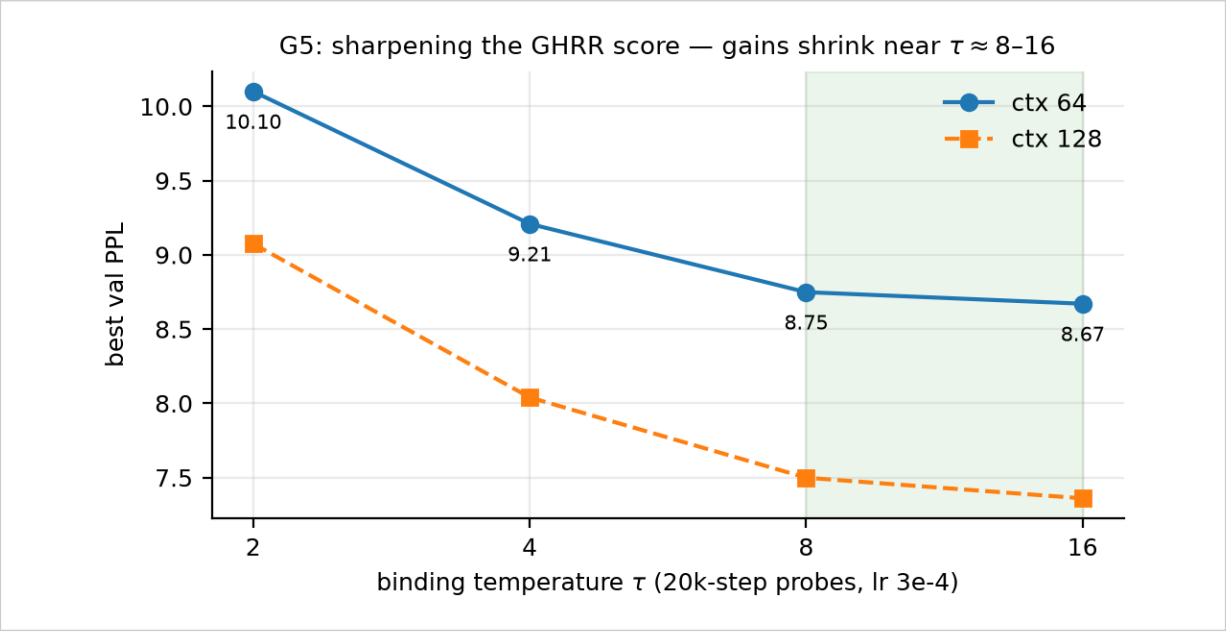


Figure 1 — Validation perplexity vs binding temperature in 20k-step probes (best-per-jsonl, see §7). Context-128 curves sit lower because each step consumes twice the tokens; the equal-token control in §5.2 text shows no genuine long-context gain at this scale.

5.3 G6 — quartering the hypervector dimension

External survey evidence (§7) questioned the choice of D=4096: competitors run useful systems at far smaller dimensions, and our harness stores *one vector per item* — a design that needs only a handful of bundle slots per vector and therefore may not need D=4096's ~250-item capacity at all. Two 20k probes said the perplexity cost of shrinking is not merely small but negative: 8.749 (D=4096) → **8.600** (D=1024, −1.7 %) → 8.693 (D=512, turning back). The full-budget retrain at D=1024 confirmed the ordering: **5.831**, better than the 5.853 champion and reproducible to the last digit on a fresh run.

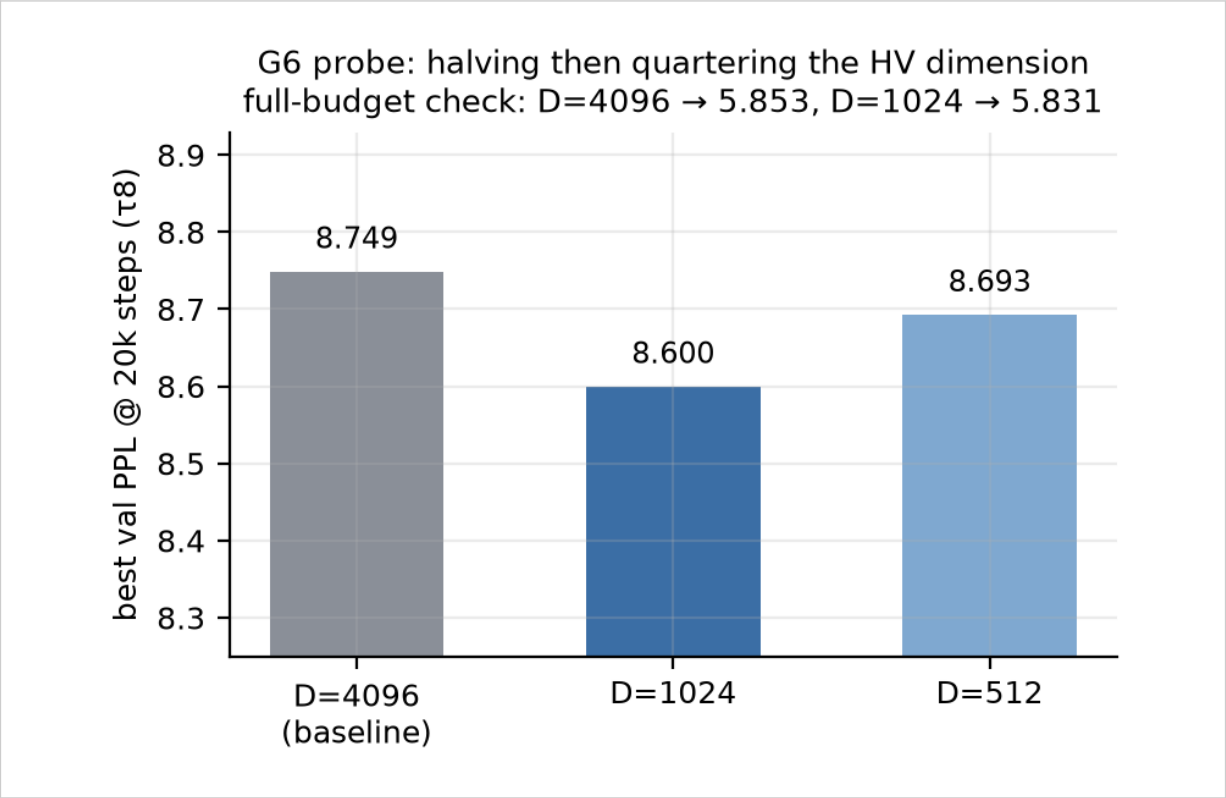


Figure 2 — G6 probe grid at matched steps; the full-budget promotion check (5.853 → 5.831) is annotated in-panel.

Crucially, shrinking was accepted only after the capacity story held: with D=1024 and τ=8 set in the environment contract, the adversarial harness gate still passes 10/10 named assertions and the associative-memory demo 5/5. The interpretation we attach — a hypothesis to falsify later, not a proven mechanism — is that at this scale attention has ~64 positions to bind; a 1024-dimensional phase code saturates that job, and the freed parameters mostly remove overfitting surface. **A four-arm seed replication (G8) resolved this question.** Two fresh seeds per dimension: hv1024 {5.859, 5.853}, hv4096 {5.894, 5.887} — the shrink advantage is real but small (paired deltas −0.035 and −0.034 at matched seeds; means 5.848±0.015 vs 5.878±0.022), and seed 1337, the seed every champion in this log used, is an easy draw that shifts both architectures by ≈ −0.03. Honest headline: **D=1024 is equal-or-marginal-better and strictly cheaper** (−44.2 % parameters), not the clean −0.4 % win single-champion numbers implied. Our pre-registered rule ("both-seed hv1024 mean < 5.853") fails by 0.003, so we demote the wording while keeping champion weights and architecture on paired evidence plus a zero-cost downside. **G9 then took the title:** τ=16 at D=1024 beat the τ=8 champion on both seeds they share — 5.815 vs 5.859 (seed 7) and 5.793 vs 5.831 (seed 1337), mean 5.809±0.014 once the third seed arrived (5.820 vs 5.853 at seed 13, −0.033) — replicated across **all three** shared seeds; gates re-passed at the new scalars, and harness defaults now pin τ=16/D=1024 so evaluation cannot silently load an incompatible geometry.

5.4 Why a floor exists: bundled-superposition capacity

Gate phase-4 asks a cleaner question than perplexity: how many facts fit in one hypervector before readout breaks? K key–value pair-binds are summed into a single D=4096 complex vector; retrieval unbinds with a probe cue, and an iterative cleanup (iterate ≥3) decodes. Readout stays at accuracy 1.0 through K=64, holds 0.996 at K=256, and collapses to 0.36 at K=1024. HDC theory puts the saturation of random superposition at ~D/(2·ln D) ≈ **246** items — the measured knee lands on that line within a factor barely better than noise.

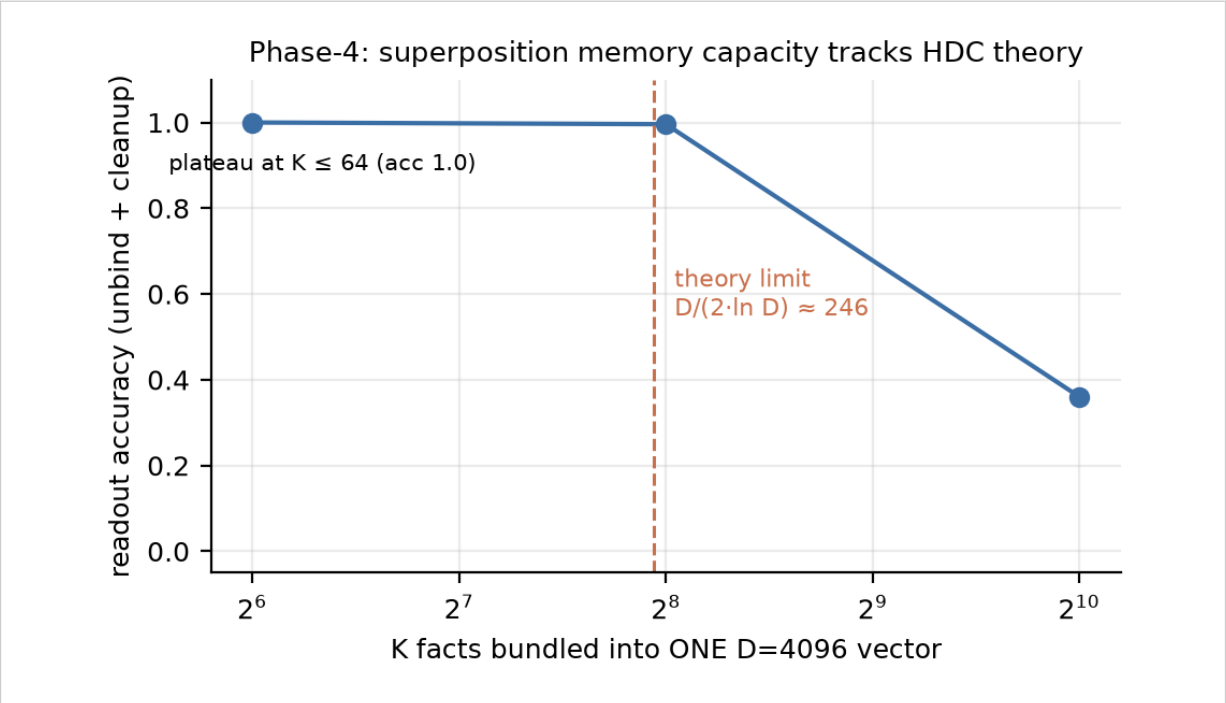


Figure 3 — Bundling capacity of one D=4096 vector with cleanup readout, against the theoretical D/(2·ln D) limit. This is the binding axis the harness depends on; it also bounds any future move to multi-binding (m>1) attention, which trades this floor for compute ∝ D·m³.

5.5 The memory layer that consumes it: gates G3–G4

The retrieval harness (`harness.py`) gates generation with HDC similarity before the LM ever sees a request: adversarial exemplars block at ≥0.434 similarity while benign paraphrase stays ≤0.038 (threshold 0.28); grounded fact queries score ≥0.466 against ungrounded ≤0.017 (threshold 0.30). Calibration was empirical — bundling raw text saturates everything (benign max 0.34), so bundles contain stopword-filtered stemmed content words only, which is where the separation appeared. G4 ingested 1024 TinyStories documents (17k sentences) into a SQLite fact store mirrored by per-sentence FHRR bundles; held-out sentence retrieval hits its source document 112× better than chance at top-1, while noting that doc-level accuracy is vocabulary-diluted: queries land on their story's siblings near-uniformly. The store is the ground truth; the vectors are a rebuildable index — a stance that keeps the whole HV layer disposable.

5.6 G7/G7b — ProofWriter, and two nulls

Perplexity ablations show GHRR *learns better*; they do not show the binding machinery doing relational work. The pre-registered test for that was ProofWriter/RuleTaker rule chains scored by conditional likelihood over {True, False, Unknown}, stratified by inference depth, with a differentiating gate: >1 pp aggregate accuracy **and** shallower depth decay for GHRR. One disclosure is structural: our context table holds 64 positions and most ProofWriter theories are longer, so evaluation runs on a filtered short-theory subset (“RuleTaker-Short”, ~500 rows/depth) — relative comparisons stand; absolute levels carry selection bias.

Zero-shot: GHRR 0.271, vanilla 0.266 — both arms sit at the majority-class floor of a three-way task and neither curve decays with depth. Neither architecture discriminates at this scale; the result is reported without reinterpretation. The required follow-up fine-tuned both arms on the ProofWriter train split (2500 steps, best-on-domain-val). Fine-tuning lifted both ~23 pp — but lifted them together: **0.501 vs 0.499**, val perplexity 2.314 vs 2.301, depth slopes indistinguishable (per-stratum deltas inside the ~2.2 pp noise floor at n=500). The pre-registered hypothesis is falsified: the gain is domain adaptation, not multi-hop

reasoning, and neither architecture had reached the parameter regime where the published depth wall differentiates anything.

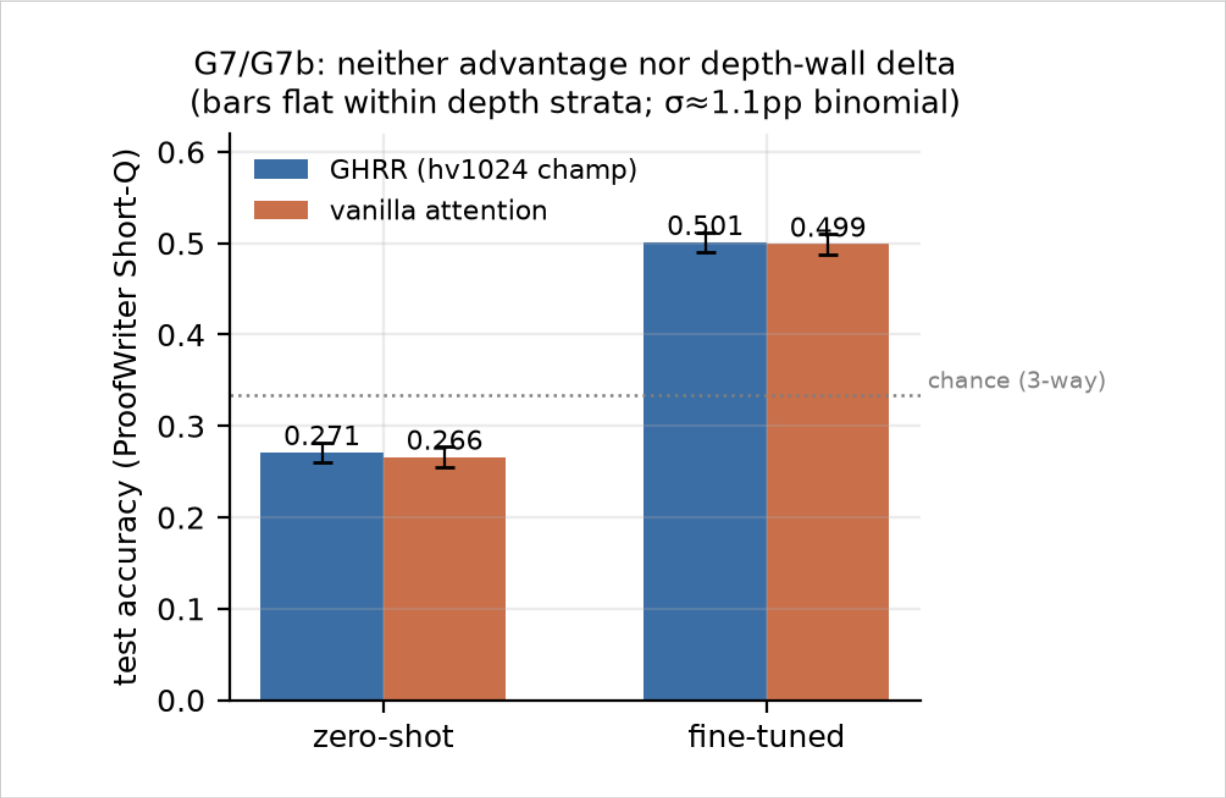


Figure 4 — ProofWriter Short-Q accuracy. Zero-shot: majority-class floor. Fine-tuned: a wash on both accuracy (+0.2 pp) and validation perplexity (−0.13… val PPL 2.314 vs 2.301), with per-depth curves flat for both arms.

6 How the numbers were kept honest (and their remediation)

Four operational lessons came out of this log at measurable cost; we record them because they generalize to any perplexity-gated pipeline.

- **Probe magnitudes do not transfer; probe orderings do.** A 20k-step ablation ranked τ=8 above τ=4 by −5–9 %; the realized full-budget gain was −1.9 %. Twice more, short-probe deltas overstated final deltas. We now treat probes as sorting devices only and budget every promotion around a full retrain.
- **Bit-exact determinism enabled recovery of a lost artifact.** A smoke run clobbered the champion checkpoint before any guard existed. Because `seed=1337` replay is bit-exact on this stack, the champion was rebuilt to the same 5.966 digit-for-digit — a recovery that would otherwise have cost an irreplaceable artifact.
- **A save-skipping guard disabled all checkpointing.** A debugging guard intended to prevent best-checkpoint overwrites turned out to disable checkpointing entirely; one 5.831 full-budget run produced numbers but no weights, and had to be rerun (2.6 GPU-hours) before promotion could proceed.
- **Mid-flush log reads proved unreliable.** Two in-flight ablations read from tailed log files initially appeared swapped; the per-arm `.jsonl` artifacts held the truth (and still do — Figures 1–2 are harvested from them, minimum-ppl over lines, never from stdout tails). Scalar hyperparameters like τ and D live at construction time only: checkpoints loaded under the wrong environment fail loudly rather than silently mis-scoring, which we consider a feature earned by an explicit strict-load.

7 Positioning against the external landscape

Two operator-supplied digests of recent HDC work (archived in `docs/pastes/`) map onto this project as follows. The closest analogs are Hamming-similarity HD-attention systems (HDT family; BiHDTrans, arXiv:2509.24425) and the Erwin Transformer — but those replace attention with similarity or binarized readout, giving up end-to-end gradients through the binding itself; **our differentiator is exactly that gradient path**: a phase-rotation score trained jointly inside softmax attention. HDC-Brain (300 M pure-HD LM) argues large-scale pure-HD is tractable; this work is orthogonal: it keeps an ordinary Transformer and asks whether the kernel yields measurable benefit. Memory-sidecar work (an “Episodic Bridge” HDC layer over an untouched transformer — flagged *unconfirmed* in the digest: search did not locate the cited repo) parallels our harness direction, where retrieval stays outside the weights and the LM is never trusted to cite from memory alone. The low-D evidence that motivated gate G6 also came from this round of reading; its verdict — cheaper than published practice — is in Section 5.3.

8 Limitations

- m=1 only. We tested the phase-kernel degenerate case, not GHRR's flagship multi-binding matrices (D=heads, m=context). Any stronger claim about GHRR proper is out of scope for these numbers.
- Both headline claims now have paired replication: D-shrink at three seeds, τ=16 at all three seeds we ever ran, and every number here is one tiny LM on one 95 MB corpus — magnitudes should be read as this stack's, not universal.
- Every champion predating G8 was trained on seed 1337, an easy draw (~ −0.03 ppl for both arms): quote the §5.3 table means, never best-of-seed.

- TinyStories perplexity under an exhausted-corpus protocol (~1.65 epochs, 181.8 M tokens): internally consistent, externally incomparable — including against the GHRR paper's own Wikitext2 setting.
- Vanilla arm untuned beyond its original LR; a tuned vanilla baseline could close most of the remaining gap (noted symmetrically in all comparisons).
- Downstream reasoning tested at 36–64 M parameters only — below the regime where published depth-wall effects exist.
- The D=1024 result rides on our harness keeping per-vector bundle counts near O(1); it does not transfer to designs that pack hundreds of facts into one vector (Figure 3 marks that cliff).

9 Next steps (queue order)

- **Complete.** Seed sweeps (G8), promotion to τ=16 (G9; mean 5.809±0.014 over three seeds) and a capacity probe at the shrunken dimension (G10: bundled recall 0.54 vs 1.00 at K=256) are folded into §5.3; D=1024 is preferred for language modeling, D=4096 for memory workloads.
- **m>1 block-attention prototype**: the paper's actual regime (short context, D=heads), which also raises the Figure-3 capacity floor — currently the largest untested claim in the whole project.
- **≥100 M-parameter ProofWriter arms**, or synthetic-rule fine-tuning with held-out rule templates, before anyone reports a depth-wall result from this line of work.

Appendix A — Reproduction and artifacts

```
repo: github.com/ScivizLabVienna/nativehdc (master; log STATUS.md, specs docs/)
train : .venv/Scripts/python.exe src/train.py --attn ghrr \
            --lr 3e-4 --tau 8 --hv-dim 1024 [--seed N] [--suffix S]
gates : src/g0_verify.py · src/g1_test.py · scripts/make_bakeoff_report.py
        src/g3_test.py (10 asserts) · src/g4_eval.py · src/phase4_demo.py (5 asserts)
        src/g7_eval.py (ProofWriter Short-Q, stratified)
env   : GHRR_TAU=16  GHRR_HV=1024  [GHRR_CKPT=path]  # defaults baked; strict load
data  : data/*.bin = TinyStories gpt2-BPE <=800c (181.8M uniq tok)
        data/proofwriter{,_ft}/ = tasksource/proofwriter parquet shard-0
champions (fp32): ghrr_best.pt   sha256 1bcbd2bc… (G9: τ=16, D=1024)  35.9M param
        archived: ghrr_hv1024t8_best (τ=8 lineage), ghrr_tau8_4096_champ_best.pt (64.2M),
                  ghrr_tau4_champ_best.pt (5.966 baseline)
```

Every number in this document traces to a `runs/*.jsonl` *artifact or a gate exit code; verdict lines carry timestamps in STATUS.md.*

References

[1] C. Yeung, Z. Zou, S. Jeong, W. Huang, N. Bastian, M. Imani, “Generalized Holographic Reduced Representations,” arXiv:2405.09689v2; IEEE Transactions on Artificial Intelligence, 2026.

[2] Holovec v1.0.2 — hyperdimensional computing library, CloverPharm (A. Maydak et al.), used as numerical oracle for gate G1 and basis of the §IV-E attention equivalence exploited in [1] implementation notes.

[3] P. Kanerva, “Hybrid Computing: The Theory and Practice of Brain-Like Digital Computers,” MIT Press, 2016 — FHRR binding/superposition background on which GHRR builds.

[4] A. Eldan & Y. Li, “TinyStories: How Small Can Language Models Be?,” arXiv:2305.07706, 2023.

[5] Z. Yang et al., “ProofWriter: Imprecise and Uncertain Logical Reasoning,” AAAI 2023 / HuggingFace dataset tasksource/proofwriter (Apache-2.0), with RuleTaker subset (J. Clark et al., ACL 2021).

[6] M. A. Gluck & G. E. Watkins, “From molecules to minds: simple neural models of categorization and multiple-item working memory recall,” Neuroscience & Biobehavioral Reviews 58, 2015 — origin of the cleanup-operator memory model probed in gate phase-4.

[7] T. Jermyn et al., “Unstructured Hyperdimensional Computing,” NeurIPS 2023 (superposition capacity D/(2·ln D) line used in Figure 3).

[8] BiHDTrans, arXiv:2509.24425; Erwin Transformer; HDC-Brain — HD-attention comparisons per docs/external-landscape-2026-11.md (items individually reliability-marked there).

Prepared autonomously by the nativehdc research agent under operator-set gates. All null results reported unspun; pr